GSE S530 Group Project: <u>Predicting COVID-19 Test Outcomes Using Symptoms and Exposure Risk Factors</u>

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Report: Predicting COVID-19 Outcomes Using Symptoms and Exposure Risk Factors

1. Summary

This study examines the key factors that influence COVID-19 test results. It focuses on identifying how symptoms, demographic characteristics, and exposure history relate to the likelihood of testing positive. The goal is to better understand which variables are most predictive in order to support data-driven clinical and operational decision-making.

2. Business Understanding

2.1 Business Content & Objectives

Our hospital is currently experiencing an influx of flu-like symptoms during a period of constrained medical resources. Similar to the early stages of the COVID-19 pandemic, we do not have the capacity to provide full COVID-level treatment to every symptomatic patient. Therefore, it is critical that we make informed decisions to distinguish patients who are most likely to be COVID-19 positive, prioritizing a balance between recall and precision in our model evaluation, in order to contain any outbreaks as efficiently as possible.

The data available includes the following symptoms: cough, fever, sore throat, shortness of breath, headache, age group, sex, and contact history. We can analyze patterns within this data and develop a model that estimates the probability of COVID-19. This will allow us to focus treatment on the right patients and prioritize high-risk patients, without exhausting medical resources.

2.2 Business Questions

1. Given a variety of patient symptoms, how accurately can we predict whether or not they will have COVID?
2. Which symptom(s) are the most telling signs that a patient has COVID?
3. Are there patient demographics that are more susceptible to getting COVID?

3. Data Understanding

3.1 Dataset Description

Variable Name	Description or Possible Values
Cough	1 - Presence of cough 0 - No presence of cough
Fever	1 - Presence of fever 0 - No presence of fever

Sore_Throat	1 - Presence of sore throat 0 - No presence of sore throat
Shortness_Of_Breath	1 - Presence of shortness of breath 0 - No presence of shortness of breach
Headache	1 - Presence of headache 0 - No presence of headache
Age_60_And_Over	Yes - Age is 60 and over, as of 2020. No - Age is less than 60, as of 2020.
Sex	Female Male
Contact	1 - Exposure to individuals with COVID-19 0 - No exposure to individuals with COVID-19
Result	Negative Positive

3.2 Exploratory Analysis and Key Insights

One of the first observations we noticed about the dataset was that only about 10% of the cases were COVID-19 positive. This is important to recognize before modeling because such a skewed class distribution can lead to biased predictions, where a model may achieve high overall accuracy by simply predicting the majority class. As a result, special attention must be given to evaluation metrics and techniques that account for class imbalance.

To better understand our variables and extract meaningful insights from the data, we implemented a Random Forest ensemble model and generated a feature importance bar chart (Figure 1). The results indicate that contact with a COVID-positive individual is the most influential predictor of test outcomes. This finding highlights the importance of carefully evaluating patient exposure history in clinical decision-making. It also underscores the need to prioritize precision in our model. Incorrectly predicting that an individual has COVID-19 when they do not (a false positive) can lead to unnecessary allocation of limited medical resources and potentially expose uninfected patients to the virus. Following exposure history, the next most important predictors identified by the model are fever, headache, and cost.

The figure indicates that variables such as Age_60_And_Above and Sex contribute little to the model's predictive performance. Based on this observation, we decided to re-run several of our models without these two variables to evaluate whether their exclusion would affect the results.

A detailed discussion of these findings is provided later in the report within each respective modeling section.

4. Data Preparation

In order to clean our large dataset, we first noticed null values in the Age_60_And_Above and Sex columns. To combat this issue, we decided to remove these observations from our dataset. We can do this using such a large dataset because the proportion of observations with null values compared to the entire set is negligible. Once working with a complete dataset, we decided to convert categorical variables to numerical. Specifically for our Sex columns, “MALE” = 1 and “FEMALE” = 0.

5. Modeling

5.1 Lasso Logistic Regression

To predict whether a patient will test positive for COVID-19, we first tried a Lasso logistic regression model. Logistic regression was used because the outcome variable is binary (COVID positive vs. negative), making it a good choice for estimating the probability of infection. To improve the model, interaction terms between predictors were included using a second-degree polynomial transformation. This allowed the model to capture relationships between symptoms and demographic factors that may jointly influence infection risk.

Because the dataset contains many possible predictors and interaction terms, Lasso regularization was used to create feature selection while fitting the model. Lasso penalizes the absolute magnitude of coefficients, shrinking less informative variables toward zero and effectively removing them from the model. This helps prevent overfitting and produces a more interpretable model by identifying the most relevant predictors. The final model retained interactions among the most important variables: cough, fever, sore throat, shortness of breath, headache, age above 60, sex, and recent contact with an infected individual.

Model hyperparameters were tuned to maximize F1 score, which balances precision and recall. This metric was chosen because both detecting COVID cases (recall) and minimizing unnecessary positive predictions (precision) are important for healthcare decision-making. We hope to catch as many cases as possible, while also conducting resource management and not overinducting patients. A class weight of 10 for positive cases and 1 for negative cases was also used to help with class imbalance and weight the model into correctly identifying infected individuals. Model performance was evaluated using 5-fold stratified cross-validation to have consistent class proportions across folds and provide a reliable estimate of out-of-sample performance.

Across the five cross-validation folds, the Lasso logistic regression model achieved the following average performance (also shown in Figure 2):

- Mean Sensitivity (Recall): 0.540
- Mean Precision: 0.599
- Mean AUC: 0.770

Sensitivity measures the proportion of actual COVID cases that the model correctly identifies. A value of 0.54 indicates that the model successfully detects approximately 54% of infected individuals. Although the value seems low it is much better then random choice as only around 10% of patients have the infection. Precision shows how many predicted positive cases are truly infected; the model's precision of about 60% indicates that a majority of flagged cases are legitimate infections. The AUC of 0.77 suggests that the model has a good ability to distinguish between infected and non-infected patients overall.

The classification threshold was set to 0.65, which shows a balance between catching COVID cases and controlling the number of predicted positives. In a healthcare setting, maximizing sensitivity alone could overwhelm hospitals with false positives and unnecessary testing. By tuning the model with an F1 score and adjusting the decision threshold, the model prioritizes identifying high-risk patients while still maintaining reasonable precision to support effective resource management.

5.2 Neural Network

A neural network classifier (MLPClassifier) was used to see whether a more flexible nonlinear model could improve predictive performance. Neural networks can capture complex relationships between predictors that can not be easily captured by linear models. The model used three hidden layers with 128, 64, and 32 neurons, ReLU activation functions, and the Adam optimization algorithm. The input variables were standardized using a StandardScaler before training.

To help with the class imbalance, balanced sample weights were used during model training so that positive COVID cases received greater importance during optimization. The model also used early stopping with a validation split to prevent overfitting by halting training when performance stopped improving. Performance was decided using 5-fold stratified cross-validation, making sure that each fold preserved the proportion of positive and negative cases.

After training the model on each fold, predicted probabilities were used to determine the optimal classification threshold. A range of thresholds between 0.2 and 0.8 was evaluated, and the threshold of 0.70 produced the highest mean F1 score across the folds.

Using this threshold, the neural network achieved the following average performance (also shown in Figure 3):

- Mean F1 Score: 0.569
- Mean Sensitivity: 0.527
- Mean Precision: 0.620
- Mean AUC: 0.769
- Mean Accuracy: 0.916

These results indicate that the neural network performs similarly to the Lasso logistic regression model in terms of overall predictive ability. While the neural network has improved precision, its sensitivity and AUC are essentially the same to the logistic model. This suggests that the nonlinear relationships captured by the neural network do not substantially improve predictive performance for this dataset.

5.3 Random Forest

As mentioned earlier, we implemented a Random Forest ensemble model to predict COVID-19 test outcomes. Random Forest is well-suited for handling correlated predictor variables and performs robustly across large datasets, making it appropriate for dynamic hospital environments where patient volume and case characteristics may fluctuate.

At the selected threshold ($t \approx 0.355$), the model achieved an AUC of 0.7695, indicating moderate discrimination ability. Recall for positive cases was 0.576, meaning the model correctly identified approximately 57.6% of COVID-positive patients. Precision was 0.895, showing strong performance in correctly identifying negative cases. Overall accuracy at this threshold was 0.8615. Output can be seen in Figure 4.

Feature importance (Gini-based) showed that contact with a COVID-positive individual was by far the strongest predictor (~0.51 importance), followed by Fever, Headache, and Cough. Demographic variables such as Sex and Age_60_And_Above contributed minimally. Overall, Random Forest provided a balanced operating point with moderate recall and high precision, making it a reasonable middle-ground model when both case detection and resource control are important.

5.4 Gradient Boosting Models

We also implemented a Histogram Gradient Boosting model, tuned using an $F\beta$ score with $\beta = 2$ to place greater emphasis on recall while still maintaining control over false positives. This

tuning choice was made with the goal of prioritizing the identification of true positive cases while maintaining reasonable precision.

At the chosen threshold (t = 0.36), the model achieved the same AUC (0.7695) as the other ensemble models, indicating comparable overall discrimination. However, its operating characteristics differed. Recall was 0.509, meaning it identified about 50.9% of positive cases, while precision was notably high at 0.967. Accuracy reached 0.9185, the highest among the three models at their selected thresholds. Output can be seen in Figure 5.

This model produced fewer false positives than Random Forest and XGBoost, making it more precision- and specificity-oriented. Permutation importance analysis revealed that Contact was once again the most influential predictor, consistent with findings from the other models. This suggests that prior exposure to an infected individual plays the most critical role in predicting COVID-19 test results within the dataset. Following Contact, several symptom-related variables, specifically, Headache and Sore Throat, took the number two and three rankings of importance. The consistency of these variables across different models strengthens confidence in their predictive value and indicates that both exposure history and symptomatic indicators are clear indicators of test results for COVID-19.

Overall, HistGradientBoosting is the most conservative model of the three. It is well-suited for hospital settings where avoiding unnecessary admissions and limiting false positives is a priority, even at the cost of slightly lower recall

6. Upsupervised Learning - Cluster Analysis

To perform analysis on patient symptoms, we conducted Hierarchical cluster analysis using Jaccard's distance and the single linkage metric to compare distances between clusters. Jaccard's distance was chosen since we wanted to emphasize the presence of symptoms in patients instead of the absence of symptoms. Since the dataset was so large, we filtered the data down to only unique combinations of the 5 symptoms, performed clustering on these combinations, and then matched the full dataset to the determined clusters. Output shown in Figure 6.

Hierarchical clustering determined that there were four main patient symptom categories:

1) The first, largest cluster can be described as patients that exhibit no cough, fever, sore throat, shortness of breath, or headache. Once matching these patients to their respective COVID-19 results, we can determine that this patient type is at a very low risk of being positive for COVID-19, as only 7% of the patients tested positive.
2) The second cluster, also the second largest, can be described as patients that tend to exhibit more than one viral respiratory symptom. With an 80% positive rate for this cluster, these patients can be labelled the most high risk for COVID-19.

3) The third cluster can be described as patients that only have a fever. Once matched to results, these patients had a 60% positive test rate, making them moderate-high risk patients.
4) The fourth cluster can be described as patients whose only symptom is a cough. These patients had a 40% positive test rate, making them moderate risk patients.

Some key takeaways from these clusters help our hospital to better understand which patient groups are at the most risk of developing COVID-19, as well as which ones naturally form distinct groups.
One interesting take-away is that patients with only a cough or fever separated into their own respective clusters, while other symptoms did not. This suggests that they not only are the most common symptoms, but patients with only one of these symptoms are less likely to have COVID-19 than when combined with one other symptom. Also, patients with a fever have a 20% higher positive rate than patients with a cough, reinforcing that patients with only a fever are at a higher risk of COVID-19.

Looking back on our predictor evaluation, we established that a headache was the second most influential symptom-based predictor of COVID-19. Headache symptoms are present in the cluster that presents a high risk of COVID-19, but instead of breaking into its own respective cluster it remains in the cluster with all other symptoms. This suggests that headaches are often seen in patients that have more than one symptom, and that this collection of symptoms is what makes these patients have a high risk of a positive COVID test result.

7. Evaluation

The first business objective we wanted to observe was: Given a variety of patient symptoms, how accurately can we predict whether or not they will have COVID?

All four models achieved AUC values of approximately 0.77, indicating good discrimination between positive and negative cases. In a medical context, AUC above 0.7 is generally considered acceptable, and 0.77 suggests the models meaningfully outperform random chance. The Lasso logistic regression, neural network, and Random Forest models achieved sensitivity between approximately 51–58%, meaning they correctly identify about half of COVID-positive patients when using calibrated thresholds. Given that only about 10% of patients in the dataset test positive, this represents a substantial improvement over baseline. Precision across models ranged from approximately 60–62% for the logistic and neural network models, indicating that a majority of patients flagged as high-risk are in fact infected, which supports efficient resource allocation.

The models have partially achieved this objective. They help predict outcomes and give probability scores that allow decisions to be based on risk, though sensitivity in the 51–58%

range means a portion of positive cases may still be missed. The AUC of 0.77 demonstrates that the models can distinguish between infected and non-infected patients with reasonable confidence.

In our second business objective we wanted to see which symptom(s) are the most telling signs that a patient has COVID?

Feature importance and coefficient analysis across all models consistently identified contact with a COVID-positive individual as the strongest predictor, with Gini importance of approximately 0.51 in the Random Forest model. This far exceeds the contribution of any other variable. Following contact history, the next most influential predictors are fever, headache, cough, and sore throat. Shortness of breath also contributes, though to a lesser extent in some models. Age_60_And_Above and Sex contributed minimally across models, which led us to drop these variables to analyze further into symptomatic variables. The exclusion of Age_60_And_Above and Sex did not affect results, confirming that symptom and exposure data are the key drivers of predictions, not demographics.

The models were able to achieve this objective. They clearly identify exposure history as the dominant factor and provide a ranked ordering of symptom importance (fever, headache, cough, sore throat) that can directly inform clinical screening protocols.

Cluster analysis additionally determined common patterns in patient symptoms. These patterns included asymptomatic, cough only, fever only, and multi-symptom. Muti-symptom proved to be the highest risk of COVID-19, fever only second, cough only third, and asymptomatic at very low risk.

In our third business objective, we wanted to determine whether certain patient demographics are more susceptible to testing positive for COVID-19.

As mentioned above, feature importance determined that the variables Age_60_And_Above and Sex contributed minimally to the model. Thus, we can not make a conclusion that certain patient demographics are most susceptible to testing positive for COVID-19.

8. Recommendation

Since all models performed well in predicting a positive test result, we will implement these models into our hospital's patient screening system to predict patient outcomes in different scenarios. First, we will use our lasso logistic regression model for its interpretive ability. Since coefficients and results are easy to interpret, this model can be used in conversations with hospital stakeholders or when describing the effect of symptoms on COVID results. Our random forest model has the best recall score, meaning it catches as many COVID cases as possible. This

will be used in situations where we want to minimize the spread of COVID, and don't mind allocating resources to false positives. Our neural network model has the best precision score, meaning it will minimize the occurrence of false positives the most. This is best for when the hospital is nearing capacity or low on necessary COVID resources because the model will still catch cases of COVID without putting unnecessary strain on the hospital.

Second, we will implement our feature analysis and hierarchical cluster analysis findings in our patient screening process as well. Patients with confirmed COVID-19 exposure will be classified as very high risk. Patients displaying multiple symptoms, particularly headache or fever, will be considered high risk. Those presenting with only a cough will be categorized as moderate risk, while asymptomatic patients will be treated as low risk.

Since we were unable to conclude differences between positive test cases for different demographics, we will continue to treat all patients between demographics as being at the same risk and not move to flag certain patient demographics.

Appendix:

Figure 1: Random Forest Feature Importance

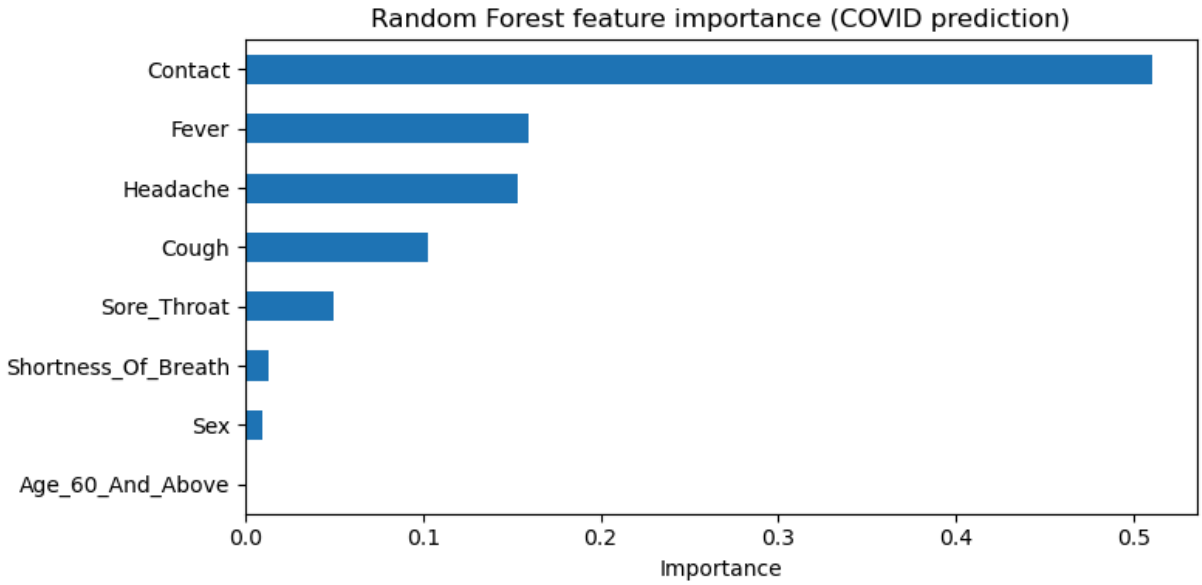

Figure 2: Final Tuned Lasso Logistic Regression Parameters List & Output Code

Model	
C	0.001
Penalty	L1
Solver	saga
Max Iterations	1000
Random State	42

Decision Settings	
Decision Threshold	0.65
Best Beta	0.5

Class Weights

Negative	1
Positive	10

Polynomial Features	
Degree	2
Interaction Only	True
Include Bias	False

Performance	
Accuracy	0.9135119003
Precision	0.5990916272
Sensitivity (Recall)	0.5399875456
F1 Score	0.5680047459

```
# 5-fold stratified CV
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=random_state)

for fold, (train_idx, test_idx) in enumerate(cv.split(X_interact, y), start=1):
   X_train, X_test = X_interact[train_idx], X_interact[test_idx]
   y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]

   model.fit(X_train, y_train)
   proba = model.predict_proba(X_test)[:, 1]
   preds = (proba >= decision_threshold).astype(int)

   sens = recall_score(y_test, preds, pos_label=1, zero_division=0)
   prec = precision_score(y_test, preds, pos_label=1, zero_division=0)
   auc = roc_auc_score(y_test, proba)

   sensitivity_scores.append(sens)
   precision_scores.append(prec)
   auc_scores.append(auc)
```

```
Mean Sensitivity: 0.5400 (+/- 0.0028)
Mean Precision:   0.5991 (+/- 0.0042)
```

```
Mean AUC:          0.7697 (+/- 0.0036)
```

Figure 3: Neural Network Code and Output

```python
# MLPClassifier: deeper (128, 64, 32), ReLU, Adam solver
model = MLPClassifier(
   hidden_layer_sizes=(128, 64, 32),
   activation='relu',
   solver='adam',
   alpha=0.001,
   batch_size='auto',
   learning_rate='adaptive',
   max_iter=200,
   random_state=42,
   early_stopping=True,
   validation_fraction=0.1,
   n_iter_no_change=10
)
```

```
--- Means (threshold=0.70, chosen by best mean F1) ---
Mean F1:           0.5692 (+/- 0.0099)
Mean Sensitivity:  0.5265 (+/- 0.0215)
Mean Precision:    0.6199 (+/- 0.0280)
Mean AUC:          0.7686 (+/- 0.0043)
Mean Accuracy:     0.9161 (+/- 0.0036)
```

Figure 4: Random Forest Tuning Code & Threshold Testing

```
from sklearn.model_selection import RandomizedSearchCV
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(random_state=321, class_weight="balanced")

param_distributions = {
    "n_estimators": [200, 300, 500, 700],
    "max_depth": [6, 8, 10, 14, None],
    "max_features": ["sqrt", "log2"],
    "min_samples_leaf": [1, 5, 10, 25, 50],
    "min_samples_split": [2, 5, 10, 25, 50],
}

search_f1 = RandomizedSearchCV(
    rf,
    param_distributions,
    n_iter=30,
    scoring="f1",
    cv=5,
    n_jobs=-1,
    random_state=321,
    verbose=1,
)
search_f1.fit(X_tune, y_tune)
best_f1 = search_f1.best_estimator_
best_f1.fit(X_train, y_train)
print("Best F1 params:", search_f1.best_params_)
```

Threshold	Recall	Precision	FP	FN
0.300	0.794	0.156	176,591	8,492
0.350	0.794	0.156	176,591	8,492
0.355	0.576	0.393	36,742	17,472
0.360	0.576	0.393	36,742	17,472
0.365	0.542	0.599	14,917	18,888
0.370	0.542	0.599	14,917	18,888
0.375	0.542	0.599	14,917	18,888
0.380	0.542	0.599	14,917	18,888
0.400	0.542	0.599	14,917	18,888

Figure 5: HistGradient Boosting Tuning Code & Threshold Testing

```
BETA = 2  # favor recall, but still cares about precision
fbeta_scorer = make_scorer(fbeta_score, beta=BETA)

gb = HistGradientBoostingClassifier(
    random_state=321,
    early_stopping=True,
    validation_fraction=0.1
)

param_distributions = {
    "learning_rate": [0.03, 0.05, 0.07, 0.1],
    "max_iter": [200, 400, 800],
    "max_depth": [3, 5, 7, 9, None],
    "max_leaf_nodes": [15, 31, 63, 127],
    "min_samples_leaf": [20, 50, 100, 200, 500],
    "l2_regularization": [0.0, 0.1, 0.5, 1.0, 5.0, 10.0],
}

search = RandomizedSearchCV(
    estimator=gb,
    param_distributions=param_distributions,
    n_iter=30,
    scoring=fbeta_scorer,
    cv=5,
    n_jobs=-1,
    random_state=321,
    verbose=1
)

search.fit(X_tune, y_tune)

print(f"Best params (F{BETA}):", search.best_params_)
print(f"Best CV score (F{BETA}):", f"{search.best_score_:.4f}")
```

Threshold	Recall	Precision	FP	FN
0.300	0.526	0.629	12,791	19,536
0.350	0.526	0.629	12,791	19,536
0.355	0.526	0.629	12,791	19,536
0.360	0.526	0.629	12,791	19,536
0.365	0.516	0.637	12,142	19,936
0.370	0.511	0.641	11,766	20,159
0.375	0.511	0.641	11,766	20,159
0.380	0.509	0.643	11,638	20,248
0.400	0.509	0.643	11,638	20,248
0.450	0.509	0.643	11,638	20,248
0.500	0.407	0.697	7,283	24,440

Figure 6: Cluster Analysis Results

```
Cluster
0    922905
1     31429
3     12202
2     11957
Name: count, dtype: int64
```

	Cough	Fever	Sore_Throat	Shortness_Of_Breath	Headache	Cluster	Result
Cluster							
0	0.000000	0.000000	0.000000	0.000000	0.000000	0.0	0.071365
1	0.470298	0.489102	0.289414	0.087149	0.609787	1.0	0.800566
2	0.000000	1.000000	0.000000	0.000000	0.000000	2.0	0.595969
3	1.000000	0.000000	0.000000	0.000000	0.000000	3.0	0.400098